PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

Health360 – Smart Healthcare Analytics: Patient Risk & Lifestyle Insights is a healthcare data analytics project designed to transform raw healthcare data into meaningful visualisations and understandable insights. The project uses the synthetic dataset smart_healthcare_dataset.csv, which contains patient-related information such as age, gender, body mass index (BMI), blood pressure, exercise level, alcohol and smoking consumption, and disease indicators including diabetes, heart disease, and stroke.

The project uses Tableau Desktop and Tableau Public to develop interactive dashboards and a Tableau Story for exploring healthcare data. The dashboard includes key performance indicator (KPI) cards, bar charts, pie charts, heatmaps, and interactive filters. These visualisations help users examine patient distribution across age groups, compare average blood pressure, explore BMI categories, and analyse recorded disease indicators.

The project also includes a Flask-based website developed using Python, HTML, and CSS to introduce the Health360 project and provide access to relevant project resources. GitHub is used to organise and maintain the project's source files and documentation.

Health360 combines data preparation, visualisation, and web development to demonstrate how healthcare data can be presented in a clear and interactive format. It is intended for educational and exploratory analysis rather than medical diagnosis.

1.2 Purpose

The main purpose of the Health360 project is to make healthcare data easier to understand and analyse by converting raw patient records into interactive visualisations and meaningful summaries. It aims to help users explore patterns in patient demographics, lifestyle factors, and recorded health conditions through a centralized analytics dashboard.

The specific objectives of the project are:

- **Analyse Patient Data:** To organise and examine patient information, including age, gender, BMI, blood pressure, and lifestyle factors.
- **Monitor Health Indicators:** To display key statistics related to diabetes, heart disease, and stroke using KPI cards and charts.
- **Identify Data Patterns:** To explore variations in recorded health indicators across age groups, BMI categories, and exercise levels.
- **Enable Interactive Analysis:** To provide filters for Age Group, Gender, and BMI Category so users can examine different subsets of the dataset.
- **Present Insights Visually:** To develop an interactive Tableau dashboard and Story that communicate findings clearly.
- **Provide Web Access:** To create a Flask-based website that introduces the project and links users to relevant project resources.
- **Support Learning and Reporting:** To demonstrate the practical application of data analytics, business intelligence, and web development in the healthcare domain.

Overall, Health360 aims to demonstrate how data visualisation can support healthcare data exploration and reporting.

2. IDEATION PHASE

2.1 Problem Statement

Ideation Phase
Define the Problem Statements

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS 1: Patient Health Analysis	A healthcare analyst	Analyze patient data and identify patterns in diseases, age groups, and health indicators	The data is difficult to understand in raw tables	There is no clear visual representation of patient health trends	Frustrated and concerned about missing important health insights
PS 2: Healthcare Data Visualization	A hospital administrator	Monitor healthcare indicators and explore patient health trends using interactive dashboards	Comparing multiple health indicators manually takes time	Patient information is spread across records and is not easy to interpret quickly	Overwhelmed and eager for a simpler way to understand healthcare data

2.2 Empathy Map Canvas

Empathize & Discover

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:

Reference: https://www.mural.co/templates/empathy-map-canvas



2.3 Brainstorming

Brainstorm & Idea Prioritization Template

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Sherin Kuriakose	Healthcare Analysis Visualizations	Data Visualization	1
2	Sherin Kuriakose	Interactive Dashboard with patient statistics and KPI cards	Dashboard Development	2
3	Sherin Kuriakose	Analysis of patient health indicators such as Diabetes, Heart Disease, and Stroke	Healthcare Analytics	3
4	Sherin Kuriakose	Patient distribution analysis based on Age Group, Gender, and BMI	Data Analysis	4

		Category		
5	Sherin Kuriakose	Heatmap to analyse Heart Disease across BMI Categories and Exercise Levels	Data Visualization	5

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	Healthcare Data Visualization using Tableau	High	High	1-High	Yes
2	Interactive Healthcare Dashboard with KPI Cards	High	High	2-High	Yes
3	Analysis of Patient Health Indicators	High	High	3-High	Yes
4	Patient Distribution Analysis by Age, Gender, and BMI	High	High	4-High	Yes
5	Heart Disease Analysis using Heatmap	High	High	5-High	Yes

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	**Stage 1: Discover & Access Data**	**Stage 2: Analyze & Visualize**	**Stage 3: Share & Take Action**
Actions *What does the customer do?*	• Access the healthcare dataset. • Open Tableau Desktop. • Connect to the CSV file.	• Explore the Tableau dashboard. • Apply filters and compare patient groups. • Analyze disease trends and health indicators.	• View the final dashboard and Story. • Share analytical findings with stakeholders. • Visit the Flask website to view project information.
Touchpoint *What part of the service do they interact with?*	• Healthcare CSV dataset. • Tableau Desktop. • GitHub repository and README.	• Tableau dashboard. • KPI cards, charts, and heatmap. • Age Group, Gender, and BMI Category filters.	• Tableau Public dashboard and Story. • Flask website. • GitHub project repository.

Customer Thought *What is the customer thinking?*	• "Is the dataset suitable for my analysis? • "Are the patient records accurate?" • "How do I connect the data to Tableau?"	• "Which chart best explains this trend?" • "How do age and BMI relate to health indicators?" • "Are these results easy to understand?"	• "What are the key findings?" • "Can I share these results easily?" • "Can stakeholders access the dashboard?"
Customer Feeling *What is the customer feeling?*	• Curious about the data. • Slightly confused during setup. • Hopeful about finding useful insights.	• Interested in the visualizations. • Engaged while exploring the filters. • Occasionally frustrated by complex charts or filters.	• Satisfied when the findings are clear. • Confident when presenting results. • Motivated to improve healthcare analysis.



Process Ownership *Who is in the lead on this?*	• Data analyst / project developer. • Responsible for data preparation and connection.	• Data analyst / project developer. • Responsible for dashboard design, calculations, and testing.	• Project developer / report presenter. • Responsible for publishing, website content, and sharing results.

Opportunities *How can we improve this stage?*	• Provide clear data-connection instructions. • Validate missing values and data types. • Improve dataset documentation.	• Use clear chart titles and labels. • Make filters easier to understand. • Add tooltips and improve dashboard layout.	• Make the website mobile-friendly. • Provide a simple user guide. • Deploy the Flask website online and verify dashboard links.

3.2 Solution Requirement

Solution Requirements (Functional & Non-functional)

Date	28 October 2026
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Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	[Specify what business data the solution should access or integrate to address the use case.]	High
2	Data Processing	[Specify what data preparation, transformation, or processing is required for the use case.]	High
3	Business Analysis	[What business calculations, KPIs, metrics, aggregations, or analysis are required to answer the use-case questions. .]	High
4	Business Insights & Analytical Outputs	[What insights, analytical outputs, dashboards, visualizations, reports, or result sets the solution should provide. .]	High
5	User Interaction & Analysis	[Specify how users should filter, compare, drill down, or explore information relevant to the use case.]	High
6	Data Update & Reporting	*[Specify how the solution should provide updated information and generate the required business outputs for the use case.]*	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system shall provide a simple, clear, and user-friendly interface. Dashboard titles, chart labels, filters, and KPI cards shall be easy to understand, including for users without advanced technical knowledge.
NFR-2	**Security**	The system shall protect sensitive patient information and restrict access to confidential data.

		Only synthetic or appropriately anonymized data shall be used in public demonstrations and repositories.
NFR-3	**Reliability**	The system shall display consistent and accurate results when the dataset and calculations are correct. It shall handle missing or invalid data appropriately and avoid misleading analytical outputs.
NFR-4	**Performance**	The system shall load dashboards and display charts within a reasonable time. Data preparation, calculations, and filtering should be efficient enough to support smooth analysis.
NFR-5	**Availability**	The Tableau dashboard shall be accessible through its published Tableau Public link when available. The Flask website shall be accessible whenever the application is running or deployed on a functioning server.
NFR-6	**Scalability**	The system should support the addition of more patient records, healthcare indicators, visualizations, and analytical features with minimal changes to the existing design.

3.3 Data Flow Diagram

Data Flow Diagram & User Stories

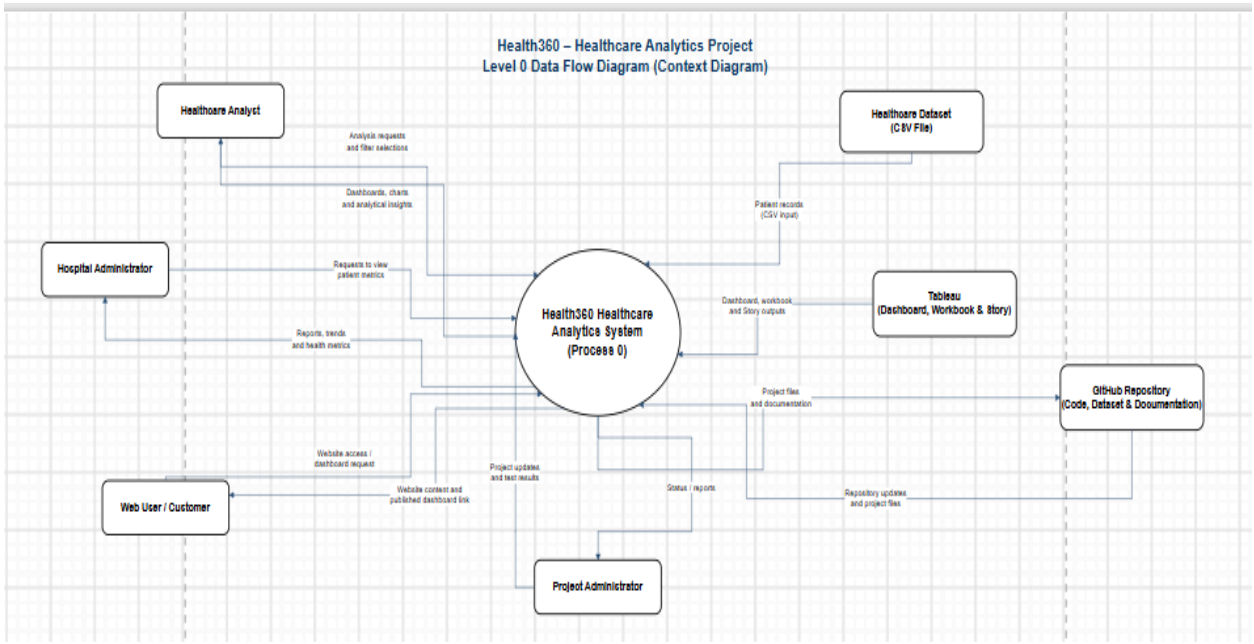
Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.


Health360 – Healthcare Analytics Project
Level 0 Data Flow Diagram (Context Diagram)
Healthcare Analyst
Healthcare Dataset
(CSV File)
Analysis requests
and filter selections
Dashboards, charts
and analytical insights
Patient records
(CSV input)
Hospital Administrator
Requests to view
patient metrics
Health360 Healthcare
Analytics System
(Process 0)
Dashboard, workbook
and Story outputs
Tableau
(Dashboard, Workbook & Story)
Reports, trends
and health metrics
Project files
and documentation
GitHub Repository
(Code, Dataset & Documentation)
Website access /
dashboard request
Project updates
and test results
Status / reports
Web User / Customer
Website content and
published dashboard link
Repository updates
and project files
Project Administrator

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Healthcare Analyst	Data Access & Integration	USN-1	As a healthcare analyst, I want to access the healthcare dataset in CSV format so that I can analyze patient information.	The CSV dataset opens and connects to Tableau successfully.	High	Sprint -1
Healthcare Analyst	Data Access & Integration	USN-2	As a healthcare analyst, I want to connect the dataset to Tableau Desktop so that I can create healthcare visualizations.	Tableau loads the required fields with appropriate data types.	High	Sprint -1
Healthcare Analyst	Data Processing	USN-3	As a healthcare analyst, I want to prepare and validate the dataset so that the analysis is accurate.	Missing values, data types, and invalid values are checked and handled appropriately.	High	Sprint -1
Healthcare Analyst	Data Processing	USN-4	As a healthcare analyst, I want to categorize patients by age group and BMI category so that I can compare different patient groups.	Age groups and BMI categories are calculated and displayed correctly.	High	Sprint -1
Healthcare Analyst	Business Analysis	USN-5	As a healthcare analyst, I want to view the total number of patients so that I can understand the size of the dataset.	The dashboard displays the correct patient count.	High	Sprint -1
Healthcare Analyst	Business Analysis	USN-6	As a healthcare analyst, I want to analyze diabetes, heart disease, and stroke cases so that I can identify disease patterns.	The dashboard displays correct case counts based on the disease indicators.	High	Sprint -1
Healthcare Analyst	Business Analysis	USN-7	As a healthcare analyst, I want to compare average blood pressure across age groups so that I can identify variations between groups.	The chart displays average blood pressure for each age group.	High	Sprint -2
Healthcare Analyst	Business Insights & Analytical Outputs	USN-8	As a healthcare analyst, I want to view charts showing patient distribution by age group so that I can understand the patient demographics.	The age-group chart displays the correct patient distribution.	High	Sprint -1
Healthcare Analyst	Business Insights & Analytical Outputs	USN-9	As a healthcare analyst, I want to analyze heart disease by BMI category and exercise level so that I can explore relationships between these factors.	The heatmap displays the relevant BMI and exercise-level categories with corresponding values.	High	Sprint -2
Healthcare Analyst	User Interaction & Analysis	USN-10	As a healthcare analyst, I want to filter the dashboard by age group so that I can focus on specific age ranges.	Selecting an age group updates the relevant visualizations.	High	Sprint -2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Healthcare Analyst	User Interaction & Analysis	USN-11	As a healthcare analyst, I want to filter the dashboard by gender and BMI category so that I can compare patient groups.	The gender and BMI filters update the relevant charts correctly.	High	Sprint -2
Healthcare Analyst	Data Update & Reporting	USN-12	As a healthcare analyst, I want to refresh the dataset when updated data is available so that the dashboard reflects the latest available information.	After the data source is updated and refreshed, the workbook displays the updated values.	Medium	Sprint -3
Hospital Administrator	Business Insights & Analytical Outputs	USN-13	As a hospital administrator, I want to view the healthcare dashboard so that I can understand key patient statistics and trends.	The published dashboard displays KPI cards and charts clearly.	High	Sprint -2
Hospital Administrator	Data Update & Reporting	USN-14	As a hospital administrator, I want to view the healthcare analysis Story so that I can understand the project findings step by step.	All published Story points open and display their intended visualizations.	Medium	Sprint -3
Customer (Web User)	Website Access	USN-15	As a web user, I want to open the Health360 website so that I can learn about the project and its healthcare analytics features.	The website loads successfully and displays the relevant project information.	High	Sprint -2
Customer (Web User)	Website Navigation	USN-16	As a web user, I want to navigate between the website pages so that I can find the information I need.	The available navigation links open the correct pages.	Medium	Sprint -2
Customer (Web User)	Dashboard Access	USN-17	As a web user, I want to access the published Tableau dashboard from the website so that I can explore the healthcare analysis.	The dashboard link opens the correct published Tableau dashboard.	High	Sprint -3
Project Administrator	Project Management	USN-18	As a project administrator, I want to maintain the project files and documentation in GitHub so that the source code, dataset, and workbook are organized.	The repository contains the relevant project files and a clear README.	High	Sprint -1
Project Administrator	Reliability & Testing	USN-19	As a project administrator, I want to test the dashboard filters, charts, Story, and website so that I can identify errors before submission.	All listed features are tested and any identified errors are documented.	High	Sprint -3
Project Administrator	Security	USN-20	As a project administrator, I want to ensure that public project files do not expose confidential patient information so that the project can be shared safely.	Only synthetic or appropriately anonymized data is shared publicly.	High	Sprint -1

3.4 Technology Stack

Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S. No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Languages used to process data, perform calculations, and build the web application.	Python, HTML, CSS	Python supports the Flask application and application logic. HTML structures web pages, while CSS styles the website. Tableau calculated fields are used for analytical calculations.
2.	Database / Data Warehouse	Stores the structured healthcare data used for analysis.	CSV file (`smart_healthcare_dataset.csv`)	CSV provides a simple format for storing patient records and importing them into Tableau. A separate database or data warehouse is not required for the current project.
3.	Data Ingestion	Loads healthcare data into the analytical environment.	Tableau Desktop, CSV data connection	Tableau Desktop connects to the CSV dataset and makes its fields available for analysis and visualization.
4.	Data Transformation and Business Logic	Prepares data and calculates the metrics required for healthcare analysis.	Tableau calculated fields, filters, aggregations, Python (Flask application logic)	Tableau supports age-group and BMI-category calculations, patient counts, average blood

				pressure, and disease analysis. Python handles the web application's logic.
5.	Data Visualization / Analytics	Creates dashboards, charts, KPIs, and analytical reports.	Tableau Desktop, Tableau Public	Tableau Desktop is used to create interactive charts, KPI cards, dashboard filters, and the healthcare Story. Tableau Public publishes the dashboard and Story online.
6.	Front-End / User Interface	Provides the interface through which users view project information and access analytical results.	Flask, HTML, CSS, Tableau Public	Flask serves the website, HTML defines its page structure, and CSS provides styling. Tableau Public provides access to the published healthcare dashboards.
	Version Control / Project Repository	Stores and organizes source code, datasets, documentation, and project files.	GitHub	GitHub keeps the project files organized and provides a repository link for sharing and submission.
	Web Application Framework	Connects the website routes with the pages and project content.	Flask (Python)	Flask provides a lightweight framework for serving the Health360 website and rendering HTML templates.
	Development Environment	Supports writing, testing, and running the application.	Python, Flask development server, web browser, Tableau Desktop	These tools allow the website to be developed and tested locally and the analytics workbook to be created and checked.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	2 Marks

Problem – Solution Fit Template:

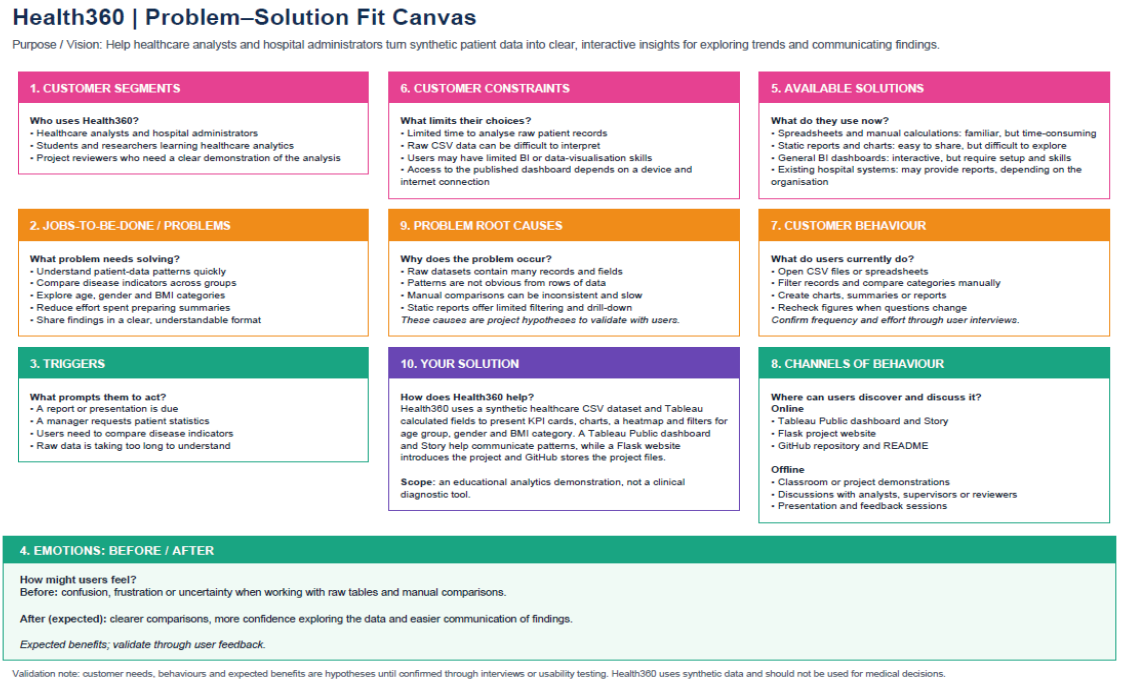
The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe



4.2 Proposed Solution

Date	15 March 2024
Team ID	EN25CS3090269
Project Title	Smart Healthcare Analytics
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop a Smart Healthcare Analytics system that transforms synthetic patient data into meaningful visualisations and insights using Tableau. The project also aims to provide a Flask-based website for presenting the project and accessing its dashboard and Story.
Scope	The project focuses on analysing patient demographics, BMI, blood pressure, exercise levels, alcohol and smoking consumption, and recorded disease indicators such as Diabetes, Heart Disease, and Stroke. It includes dataset preparation, interactive Tableau dashboards, a Tableau Story, and a Flask website. The project is intended for educational and exploratory analysis, not clinical diagnosis.
Problem Statement	
Description	Healthcare datasets contain large amounts of patient information, which can be difficult to understand when presented only in raw tabular format. Analysing patient demographics, lifestyle factors, and health indicators manually can make it difficult to identify patterns and compare different categories. Therefore, a visual analytics solution is needed to present healthcare information in a clear and organised manner.
Impact	Without proper visualisation, understanding patient distributions and comparing health indicators can be time-consuming. A centralised interactive dashboard can make data exploration easier, improve the presentation of information, and help users identify patterns within the dataset.

Proposed Solution	
Approach	The project uses a healthcare dataset containing 5,000 patient records and 21 fields. The data is connected to Tableau Desktop, where calculated fields, aggregations, charts, KPI cards, heatmaps, and interactive filters are used to analyse and present healthcare information. Tableau Public is used to publish the dashboard and Story. A Flask website developed using Python, HTML, and CSS provides project information and access to the visualisations.
Key Features	1. Interactive healthcare analytics dashboard. 2. KPI cards for Total Patients, Diabetes, Heart Disease, and Stroke. 3. Patient distribution analysis by age group and gender. 4. Average blood pressure comparison across age groups. 5. BMI and heart disease analysis using charts and heatmaps. 6. Exercise level distribution visualisation. 7. Filters for Age Group, Gender, and BMI Category. 8. Tableau Story for presenting healthcare observations. 9. Flask-based website with Home, About, Dashboard, Story, and Contact sections. 10. GitHub repository for source code and project documentation.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Computer or laptop for data analysis, Tableau development, and Flask website development.	Standard personal computer/laptop with a modern multi-core processor. No dedicated GPU required.
Memory	RAM required to run Tableau Desktop, Python, Flask, and a web browser.	8 GB RAM recommended.

Storage	Storage for the dataset, Tableau workbook, website files, and project documentation.	Available local disk space; approximately 1 GB or more is a practical allocation.
Software		
Frameworks	Framework used for developing the website.	Python Flask
Libraries	Software libraries used for website development, if required.	Flask and its required dependencies.
Development Environment	Used to create healthcare charts, dashboards, and Stories.	Tableau Desktop and Tableau Public
Frontend Technologies	Used to design and develop the website interface.	HTML and CSS
Development Environment	Used to write, run, and manage project files.	Python environment, Visual Studio Code or another code editor, and web browser.
Version Control	Used to store and manage source code and project documentation.	Git and GitHub
Data		
Data	Healthcare dataset used for analysis and visualisation.	smart_healthcare_dataset.csv
Data Attributes	Patient demographics, lifestyle information, and health indicators.	Age, Age Group, Gender, BMI, BMI Category, Blood Pressure, Exercise Level, Alcohol, Smoking, Diabetes, Heart Disease, and Stroke.
Data Usage	Used to create calculated fields, analyse health indicators, and develop interactive visualisations.	Tableau Desktop

4.3 Solution Architecture

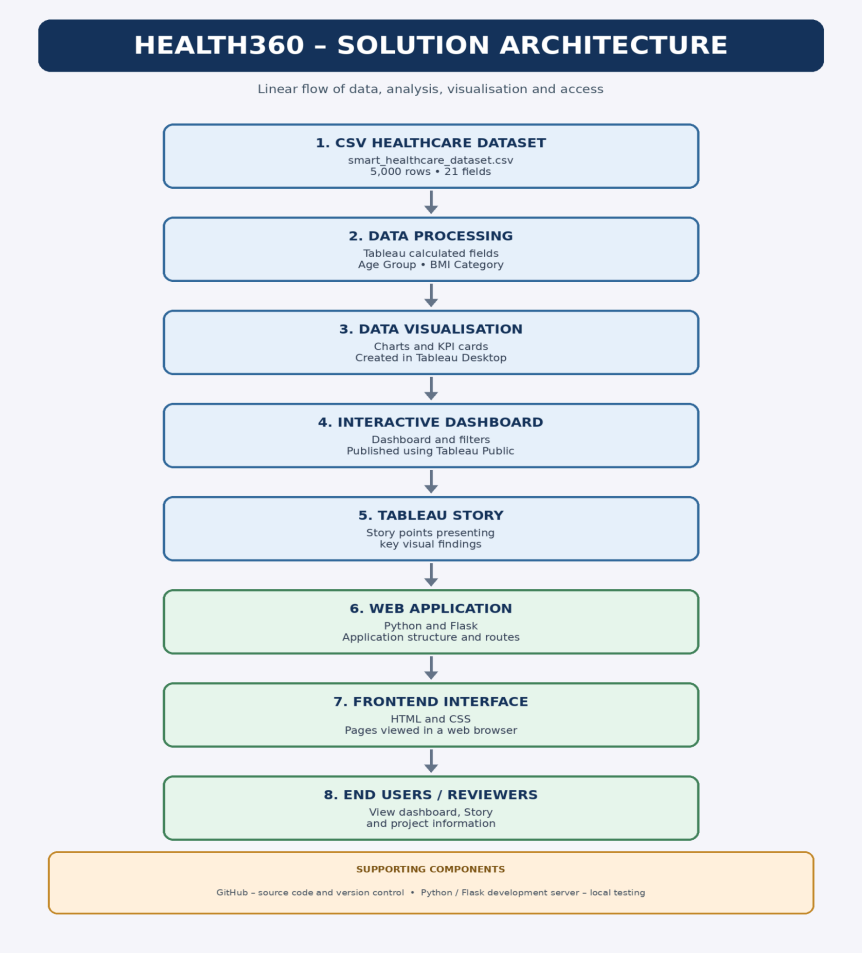
Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Healthcare Dataset	Stores patient information such as Age, Gender, BMI, Blood Pressure, Exercise Level, Alcohol, Smoking, Diabetes, Heart Disease, and Stroke.	CSV
Data Processing	Handles data preparation, calculated fields, categorisation, and aggregation required for healthcare analysis.	Tableau Calculated Fields
Data Visualization	Creates charts, graphs, KPI cards, and heatmaps to represent healthcare information visually.	Tableau Desktop
Interactive Dashboard	Displays patient statistics and healthcare visualisations with filters for Age Group, Gender, and BMI Category.	Tableau Desktop / Tableau Public
Tableau Story	Organises visualisations into a sequence to present healthcare observations and findings.	Tableau Desktop / Tableau Public
Web Application	Provides a website with Home, About, Dashboard, Story, and Contact sections to introduce the Health360 project.	Python, Flask
Frontend Interface	Designs the website layout, navigation menu, headings, buttons, and visual elements.	HTML, CSS
Version Control	Stores and manages project source code, dataset files, and documentation.	GitHub
Development Environment	Provides the environment for writing, running, and testing the Flask application.	Python, Flask Development Server, Web Browser

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Data Integration	USN-1	As a project developer, I can import the healthcare CSV dataset into Tableau for analysis.	3	High
Sprint-1	Data Preprocessing	USN-2	As an analyst, I can prepare data fields and create Age Group and BMI Category classifications.	5	High
Sprint-1	Healthcare KPIs	USN-3	As an analyst, I can display total patients and disease-related KPI cards.	5	High
Sprint-1	Dashboard Visualisation	USN-4	As an analyst, I can visualise patient distribution across age groups.	3	High
Sprint-1	Testing	USN-5	As a developer, I can verify that the dataset and initial visualisations display correctly.	4	Medium
Sprint-2	Health Analytics	USN-6	As an analyst, I can compare average blood pressure across age groups.	3	High
Sprint-2	Interactive Dashboard	USN-7	As a dashboard user, I can filter results by Age Group, Gender, and BMI Category.	5	High
Sprint-2	Health Analytics	USN-8	As an analyst, I can explore heart disease across BMI categories and exercise levels using a heatmap.	5	High
Sprint-2	Dashboard Design	USN-9	As a dashboard user, I can view healthcare KPIs and charts in a single dashboard.	4	High
Sprint-2	Tableau Story	USN-10	As a presenter, I can arrange visualisations into a Story to communicate findings.	3	Medium
Sprint-3	Web Development	USN-11	As a visitor, I can access a Flask website that introduces the Health360 project.	5	High
Sprint-3	Website Integration	USN-12	As a visitor, I can access the published Tableau dashboard from the project website, if configured.	3	High

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-3	Documentation	USN-13	As a project reviewer, I can read the README describing the project, dataset, tools, and setup.	3	Medium
Sprint-3	Version Control	USN-14	As a developer, I can store and organise project files in the GitHub repository.	3	Medium
Sprint-3	Final Testing	USN-15	As a project team, we can check dashboard filters, visualisations, Story navigation, and website links.	6	High

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	16 Sep 2026	21 Sep 2026	20	21 Sep 2026
Sprint-2	20	6 Days	22 Sep 2026	27 Sep 2026	20	27 Sep 2026
Sprint-3	20	6 Days	28 Sep 2026	03 Oct 2026	20	03 Oct 2026

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

AV = 20/6 = 3.333 points per day

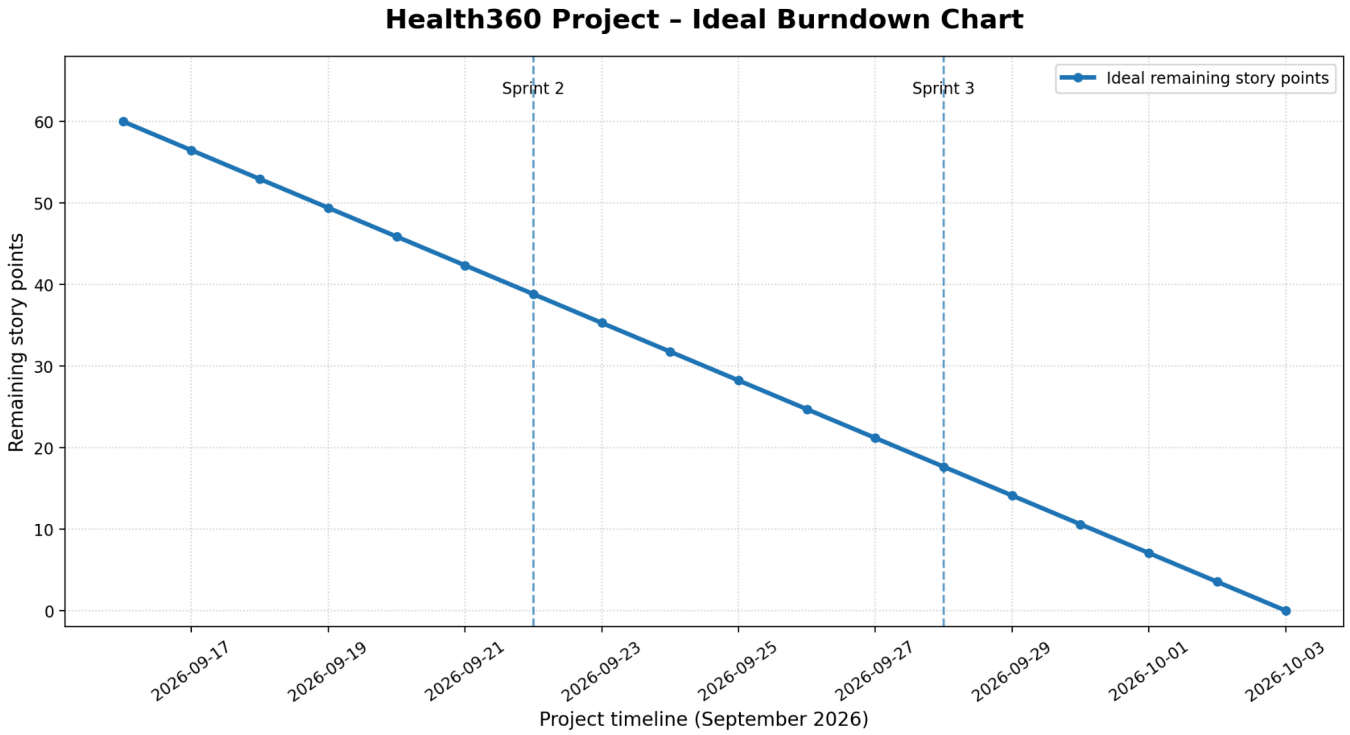
Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/
https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:
https://www.atlassian.com/agile/project-management
https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics
https://www.atlassian.com/agile/tutorials/sprints
https://www.atlassian.com/agile/project-management/estimation
https://www.atlassian.com/agile/tutorials/burndown-charts



Planned baseline: 60 story points across 3 sprints. Ideal trend only; actual daily progress was not provided.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	

Project Performance Testing:

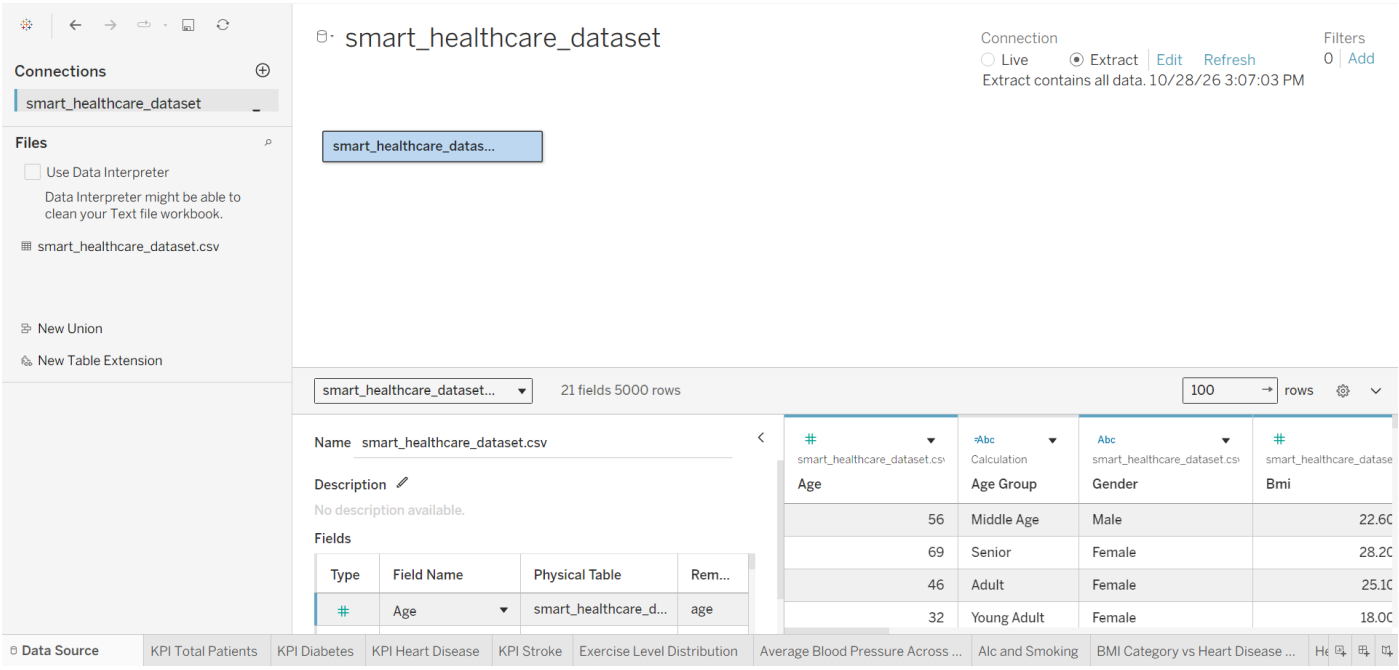
The project team shall fill the following information in the model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The smart_healthcare_dataset.csv synthetic healthcare dataset was imported into Tableau Desktop. Patient records and fields such as Age, Gender, BMI, Blood Pressure, Diabetes, Heart Disease, Stroke, and Exercise Level are used for analysis. Evidence: Screenshot of the Tableau Data Source page and a worksheet displaying data.
2.	Data Preprocessing	The Age calculated field was created, and Age Group and BMI Category are used for categorisation. Diabetes, Heart Disease, and Stroke are coded as 0 and 1. Exercise Level uses categories coded as 0, 1, and 2.
3.	Utilization of Filters	The dashboard includes Age Group, Gender, and BMI Category filters to allow users to explore and compare patient data.
4.	Calculation fields Used	Age, Age Group, and BMI Category, along with aggregations such as patient counts and average Blood Pressure. Other calculated fields should be listed if they appear in the final workbook.
5.	Dashboard design	No of Visualizations / Graphs - 9 Includes KPI cards for Total Patients, Diabetes, Heart Disease, and Stroke, along with charts such as Patient Distribution by Age Group, Average Blood Pressure Across Age Groups, a Heart Disease by BMI Category and Exercise Level heatmap, and an Exercise Level chart.
6	Story Design	No of Visualizations / Graphs - 8 Exercise level distribution, Average blood pressure across age groups, Alcohol and smoking, BMI category across heart disease, Heart disease vs disease prevalence, Patient distribution by age group, Heart disease rate by BMI and exercise level, Age group distribution by Fatigue condition

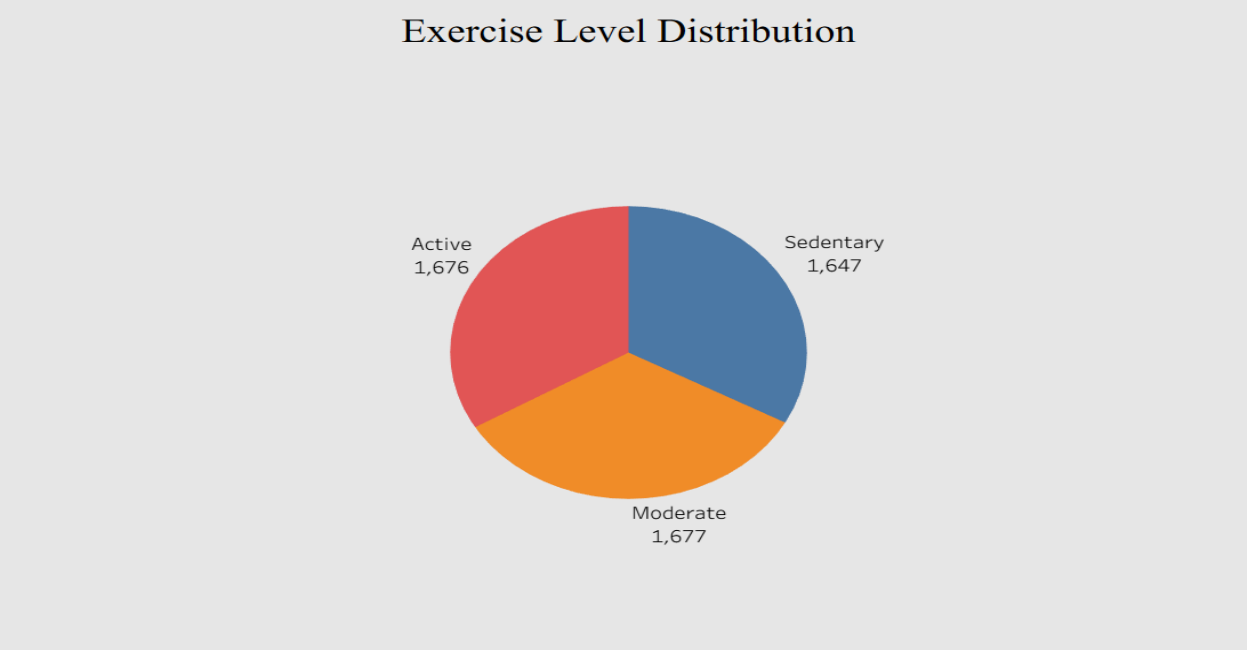
7. RESULTS

7.1 Output Screenshots

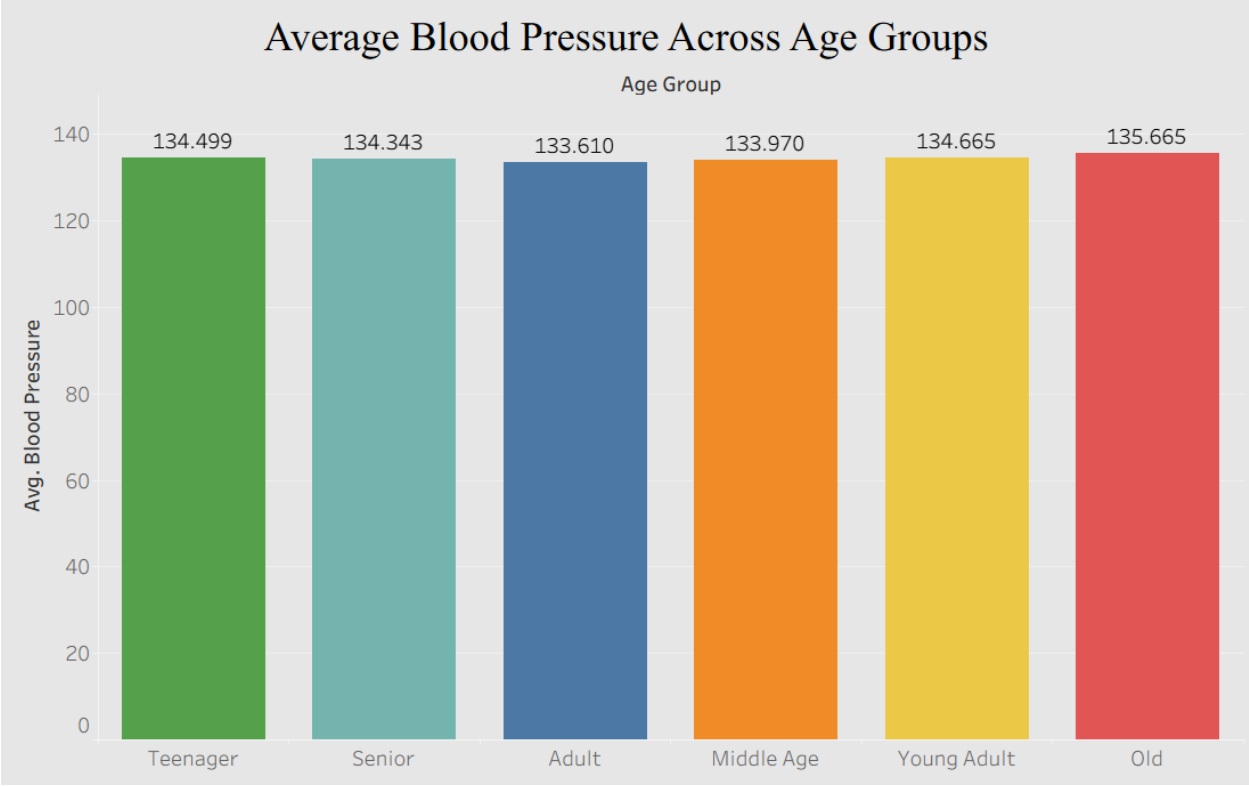
1. Tableau Data Source page showing the connected healthcare dataset



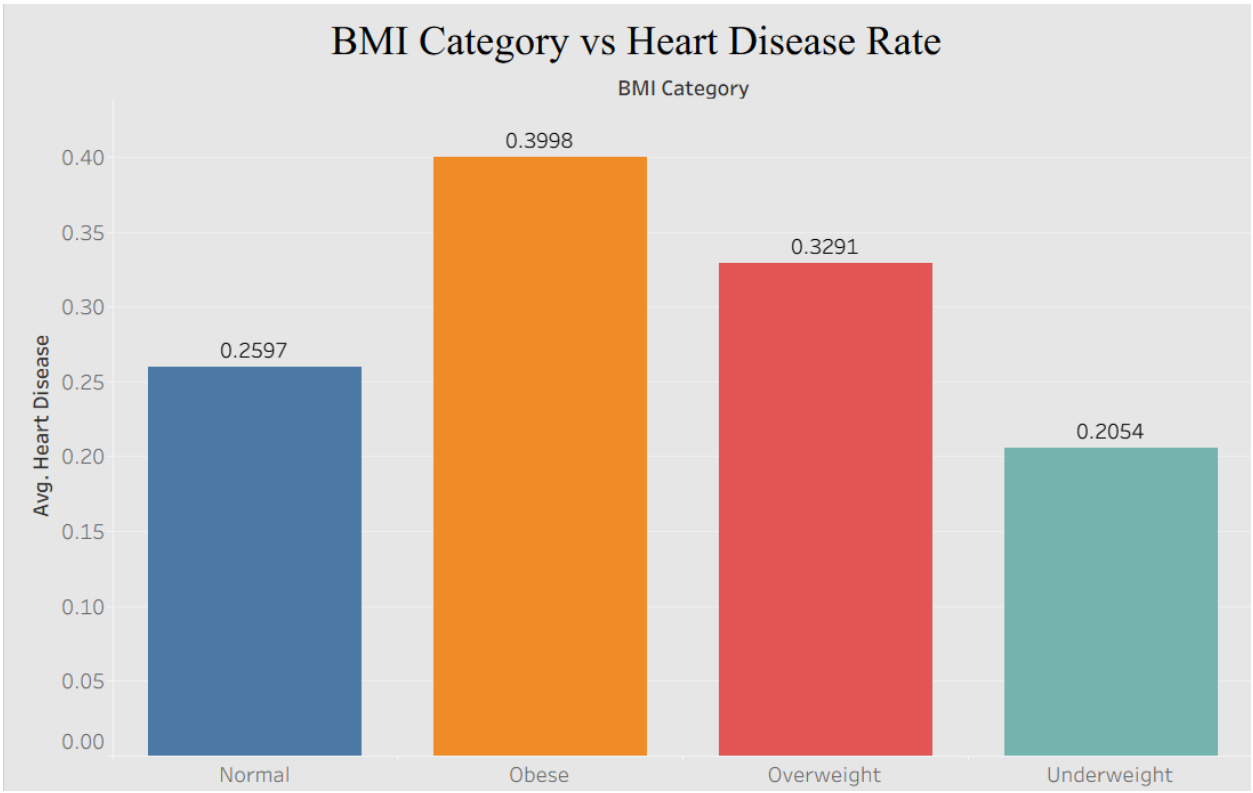
2. Exercise Level Distribution chart.



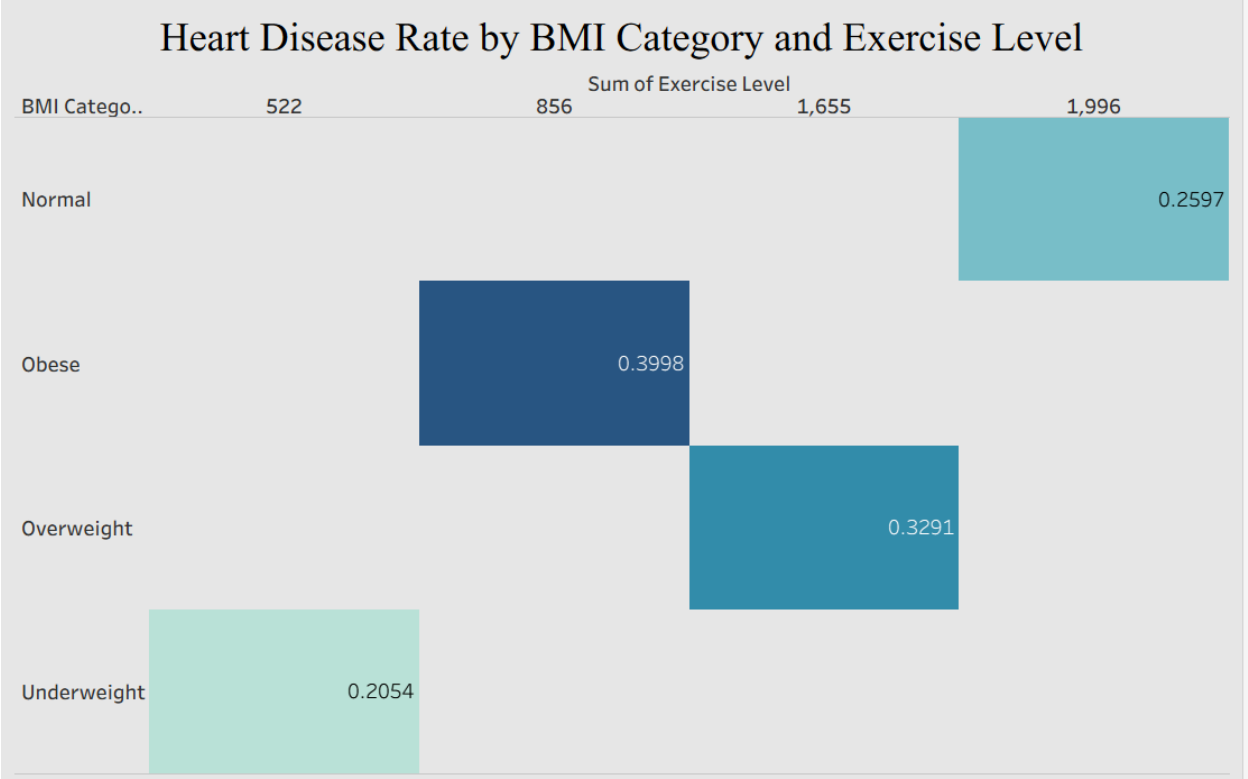
3. Average Blood Pressure Across Age Groups chart.



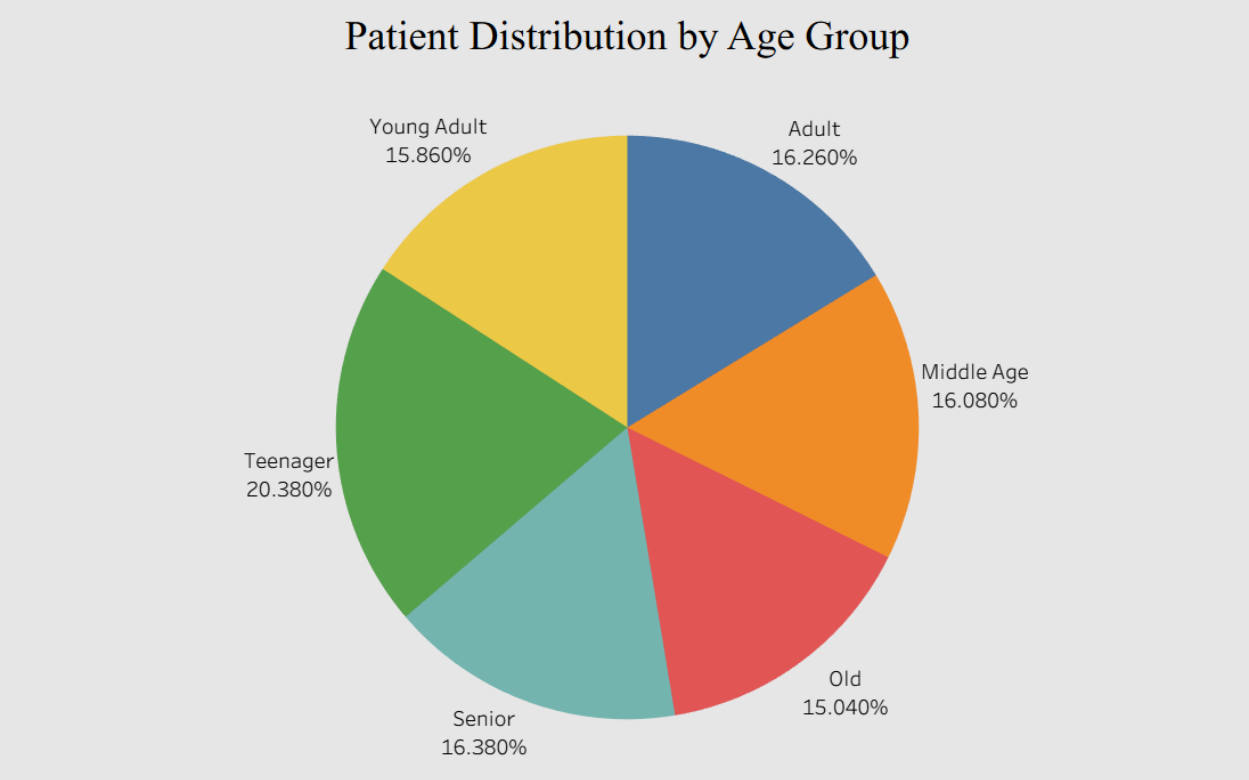
4. BMI Category vs Heart Disease Rate chart.



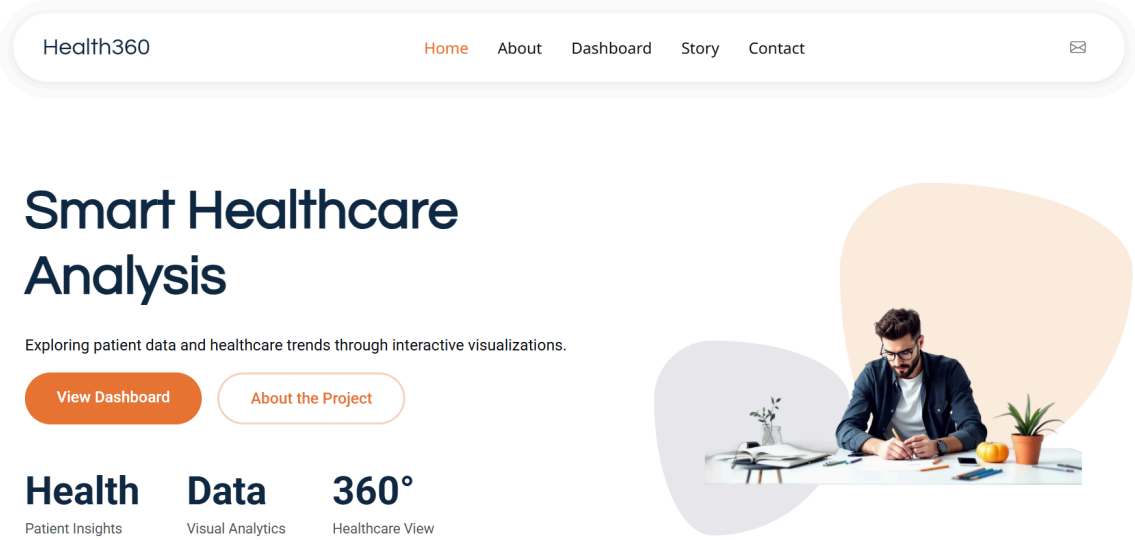
5. Heart Disease by BMI Category and Exercise Level heatmap.



6. Patient Distribution by Age Group chart.



7. Flask website homepage.



8. Dashboard Design

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	5 Marks

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities

- Responsive Design
- Custom Visuals and Icons
- Use of Infographics



Here are five potential outcomes from the dashboard:

- **Overall Patient and Disease Overview:** The dashboard represents 5,000 patients, with 1,746 recorded diabetes cases, 1,499 heart disease cases, and 1,243 stroke cases. These KPI cards provide a quick overview of the dataset and its recorded disease indicators.

- **Exercise Level Distribution:** The dataset contains 1,677 patients in the moderate exercise category, 1,676 in the active category, and 1,647 in the sedentary category. The distribution is relatively balanced across the three exercise levels.

- **Average Blood Pressure Across Age Groups:** The average blood pressure values range from approximately 133.61 to 135.67 across the six age groups. The Old group has the highest average (135.665), while the Adult group has the lowest (133.610).

- **Heart Disease Rate by BMI Category:** The Obese category has the highest average heart disease indicator at 0.3998 (39.98%), followed by Overweight at 0.3291 (32.91%), Normal at 0.2597 (25.97%), and Underweight at 0.2054 (20.54%).

- **Alcohol and Smoking Consumption Across Age Groups:** The chart compares the displayed alcohol and smoking values across age groups. The highest values appear in the Teenager group, at 514 for alcohol and 526 for smoking, while the lowest alcohol value is 358 in the Old group.

- **Heart Disease and Diabetes Prevalence:** The comparison chart shows higher diabetes counts than heart disease counts in each BMI category. The Normal category has 660 diabetes cases and 522 heart disease cases, while the Obese category has 309 diabetes cases and 321 heart disease cases.

- **Patient Distribution by Age Group:** Teenagers represent the largest displayed age group, with 20.38% of patients, followed by Seniors at 16.38%, Young Adults at 15.86%, Adults at 16.26%, Middle Age at 16.08%, and the Old group at 15.04%.

- **Age Group Distribution by Fatigue Condition:** The displayed counts vary across age groups. Teenagers have the highest count at 1,019, followed by Seniors at 819, Adults at 813, Middle Age at 804, Young Adults at 793, and the Old group at 752.

- **Interactive Dashboard Filtering:** The dashboard includes filters for Exercise Level, Gender, Age Group, and BMI Category. These allow users to explore subsets of the dataset and compare the displayed healthcare indicators.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

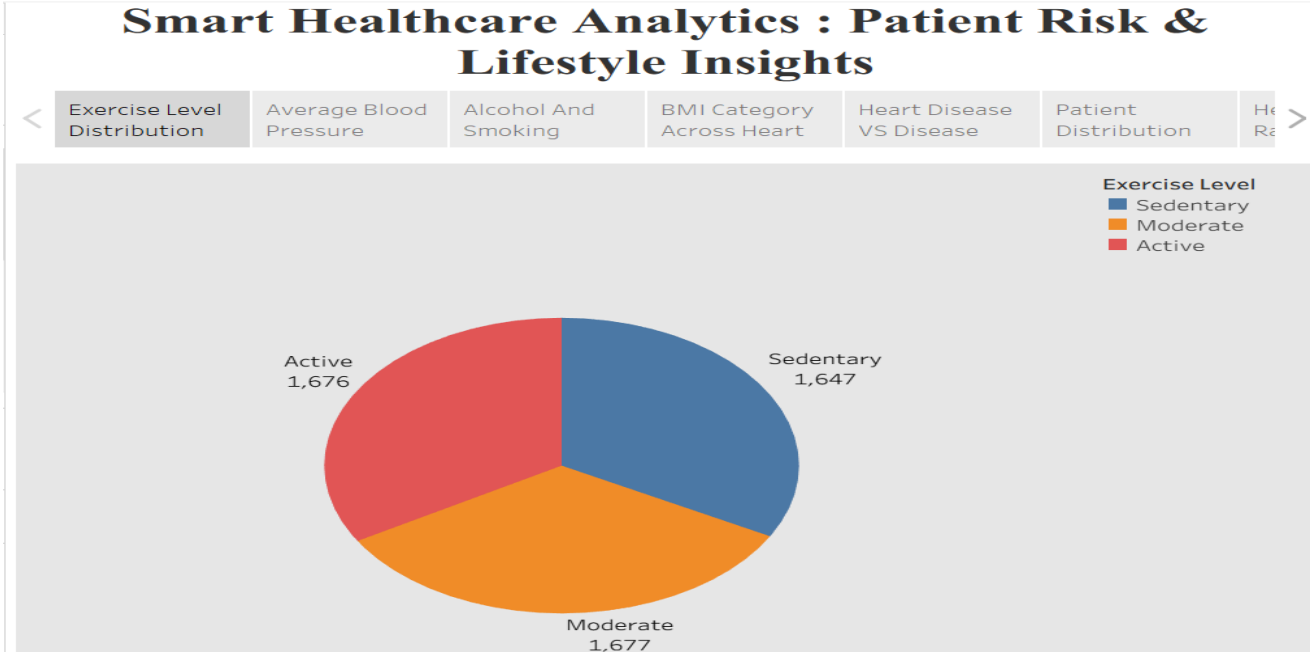
https://public.tableau.com/app/profile/sherin.kuriakose/viz/Health360Project-FINAL/Dashboard1

9. Story

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	5 Marks

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.



Observations :

- **Balanced Exercise Level Distribution:** The Moderate exercise category contains the highest number of patients at 1,677, closely followed by Active at 1,676 and Sedentary at 1,647.
- **Moderate Exercise Is the Largest Category:** The three exercise groups have relatively similar patient counts, indicating a balanced distribution of exercise levels in the dataset.
- **Small Difference Between Active and Moderate Groups:** Only one patient separates the Active and Moderate categories, showing that their recorded counts are almost identical.
- **Sedentary Group Has the Lowest Count:** The Sedentary category contains 1,647 patients, which is 30 fewer than the Moderate category.
- **Interactive Story Navigation:** The Tableau Story presents multiple healthcare visualisations through navigation tabs, allowing users to explore exercise distribution, average blood pressure, alcohol and smoking consumption, BMI-related indicators, and patient demographics.

Activity 2:

Publish Story on Tableau Public and Paste Public link below :

https://public.tableau.com/app/profile/sherin.kuriakose/viz/HealthcarePerformanceStory/Story1?publish=yes

Business Question and Visualization Report

Date	28 October 2026
Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

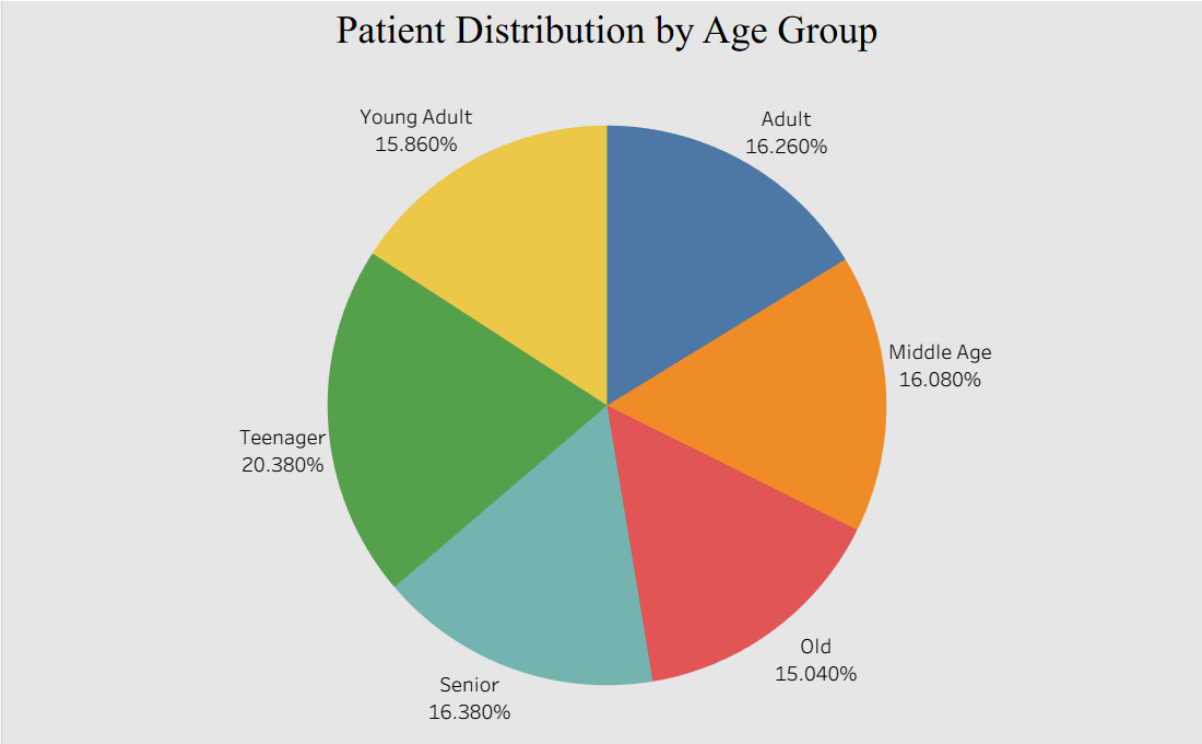
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

1. **What are the monthly sales trends?**
   - *Visualization*: Line chart showing monthly sales trends.
   - *Screenshot of visualisation*
2. **Which products have the highest revenue?**
   - *Visualization*: Bar chart comparing revenue by product
   - *Screenshot of visualisation*
3. **How is customer satisfaction varying across regions?**
   - *Visualization*: Map visualization showing customer satisfaction by region.
   - *Screenshot of visualisation*

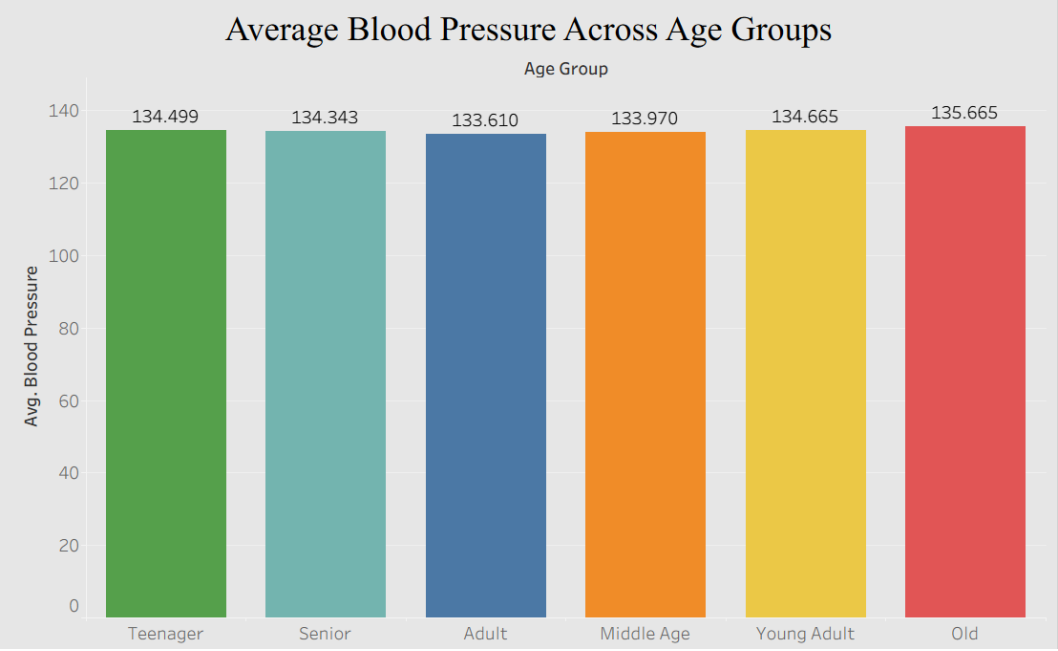
1. How are patients distributed across different age groups?

- Visualization: Bar chart showing the number of patients in each age group.
- Purpose: To understand the distribution of patients across age categories and identify groups with higher patient counts.
- Screenshot of visualisation:



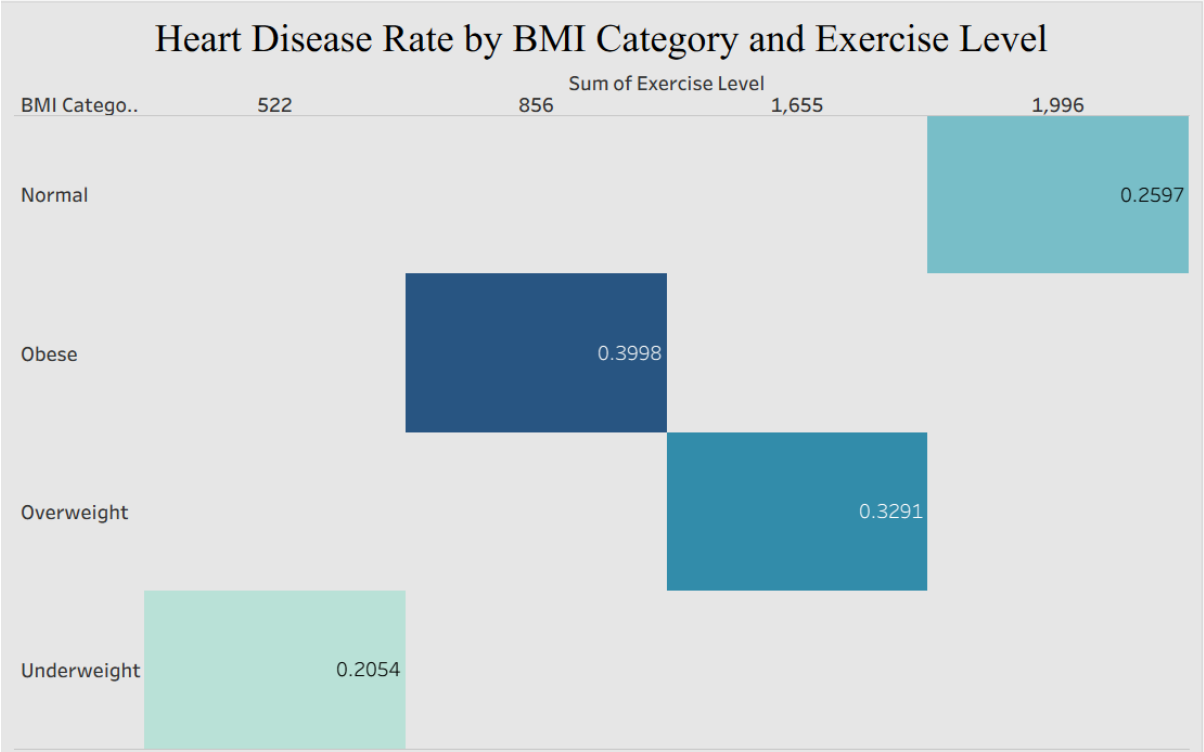
2. How does average blood pressure vary across age groups?

- Visualization: Bar chart showing average blood pressure for each age group.
- Purpose: To compare average blood pressure across different age categories.
- Screenshot of visualisation:



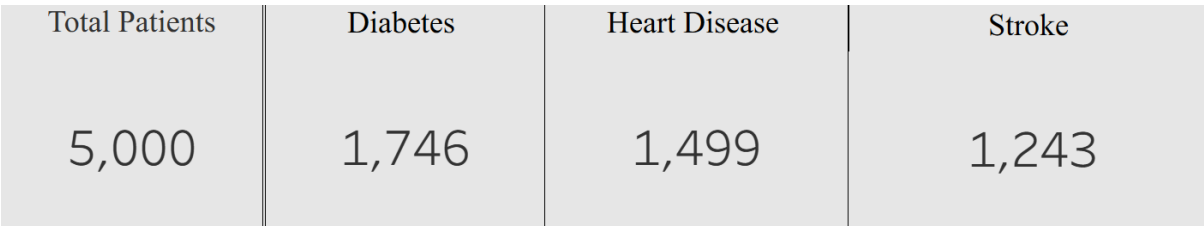
3. How does heart disease vary across BMI categories and exercise levels?

- Visualization: Heatmap comparing heart disease indicators across BMI categories and exercise levels.
- Purpose: To explore differences in the recorded heart disease indicators between BMI and exercise categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Heart Disease by BMI Category and Exercise Level.



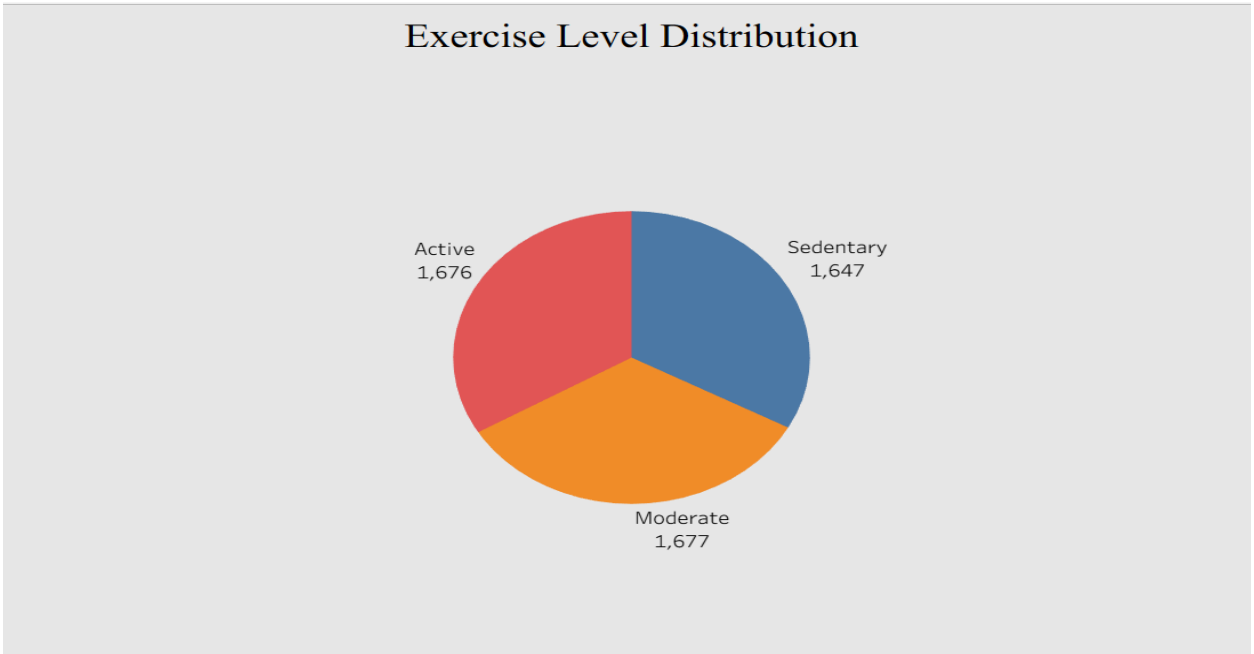
4. What is the overall number of patients and recorded disease cases?

- Visualization: KPI cards showing Total Patients, Diabetes Cases, Heart Disease Cases, and Stroke Cases.
- Purpose: To provide a quick overview of the patient dataset and the recorded disease indicators.
- Screenshot of visualisation:



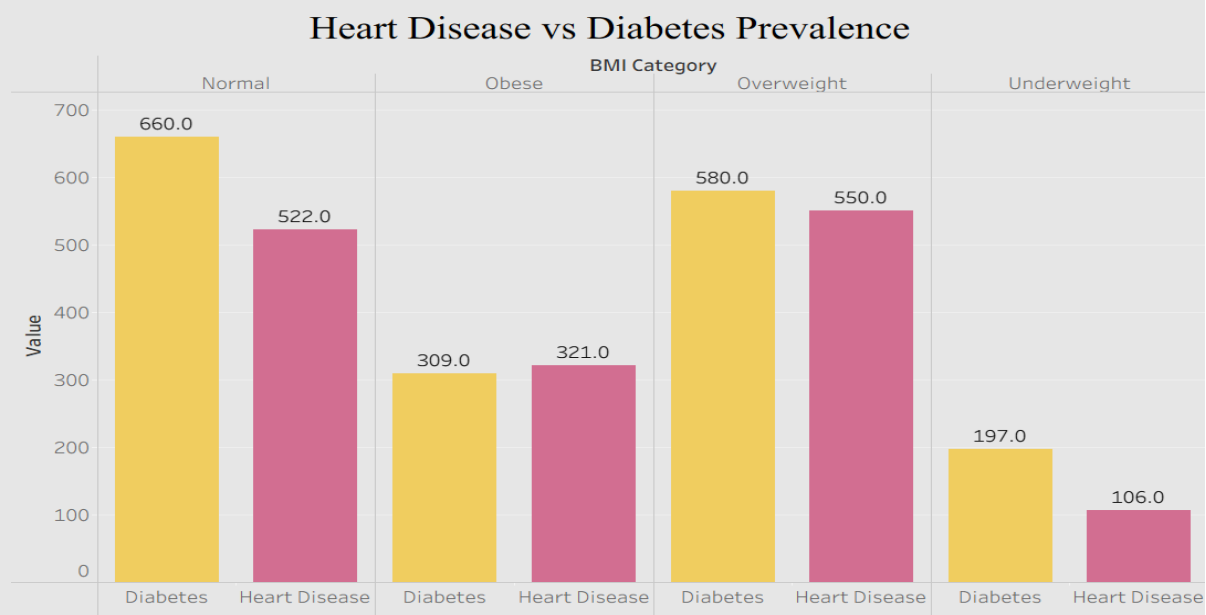
5. How does the distribution of exercise levels vary among patients?

- Visualization: Pie chart or donut chart showing the distribution of patients across Sedentary, Moderate, and Active exercise levels.
- Purpose: To understand patients' exercise habits and identify the most common activity level.
- Screenshot of visualisation:



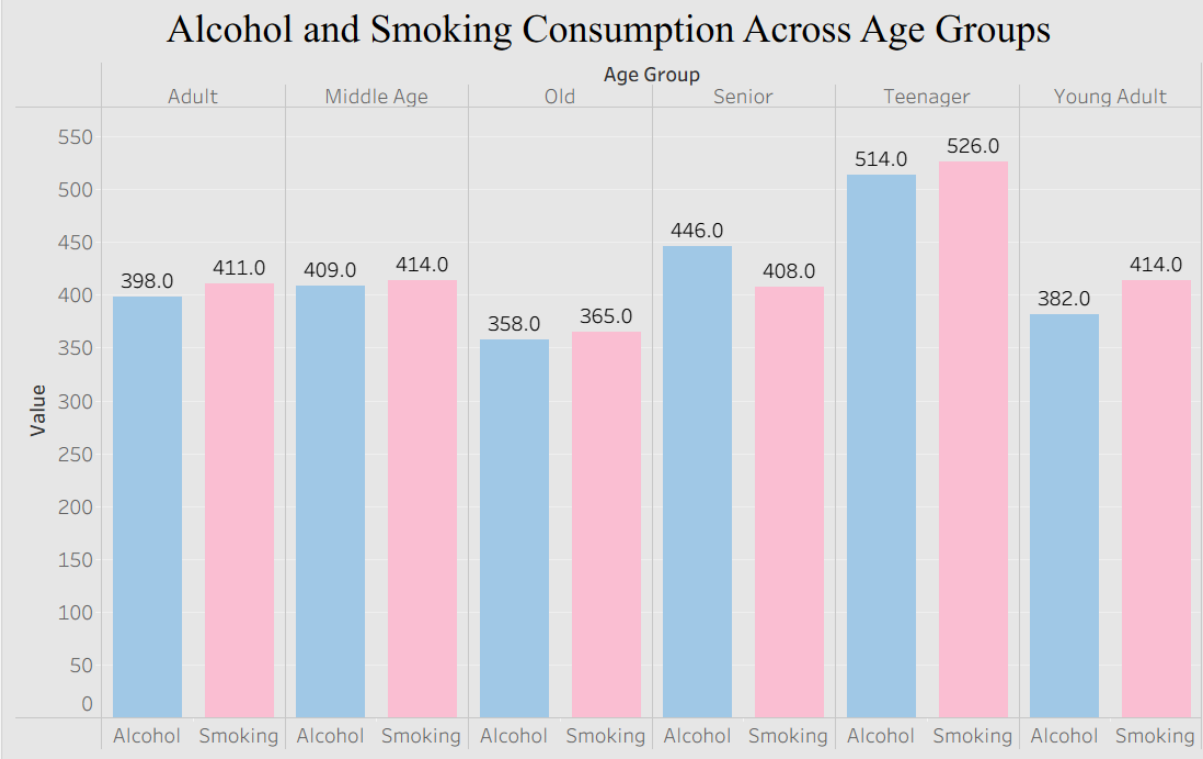
6. How do diabetes and heart disease cases compare across BMI categories?

- Visualization: Side-by-side bar chart comparing diabetes and heart disease cases across BMI categories.
- Purpose: To compare the recorded diabetes and heart disease indicators among patients in different BMI categories.
- Screenshot of visualisation:



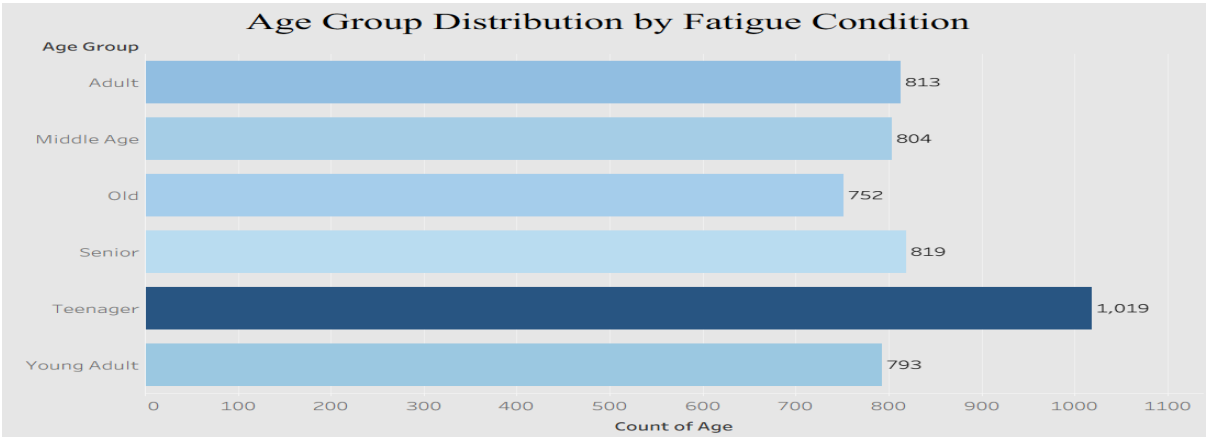
7. How do alcohol consumption and smoking indicators vary across age groups?

- Visualization: Bar chart comparing alcohol and smoking indicators across different age groups.
- Purpose: To examine how the recorded alcohol and smoking indicators are distributed among different age categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Alcohol and Smoking by Age Group.



8. How does fatigue vary across different age groups?

- Visualization: Bar chart showing the number of patients with fatigue in each age group.
- Purpose: To identify age groups with higher recorded fatigue counts and compare fatigue distribution across age categories.
- Screenshot of visualisation: Insert your Tableau worksheet screenshot for Fatigue by Age Group.



8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

1. **Easy Data Understanding:** Health360 converts raw healthcare data into charts, graphs, heatmaps, and KPI cards, making information easier to understand.
2. **Interactive Dashboard:** Tableau filters allow users to explore data based on Age Group, Gender, and BMI Category.
3. **Quick Performance Overview:** KPI cards display the total number of patients and recorded cases of diabetes, heart disease, and stroke.
4. **Identification of Patterns:** Visualisations help users compare patient demographics, exercise levels, blood pressure, BMI categories, and disease indicators.
5. **Centralised Visualisation:** Multiple healthcare charts are presented within a single dashboard for convenient analysis.
6. **Improved Presentation:** The Tableau Story helps organise visualisations and communicate findings in a structured manner.
7. **Web-Based Project Access:** The Flask website provides a platform to introduce the project and connect users with relevant project resources.
8. **Use of Open-Source Technologies:** Python, Flask, HTML, CSS, and GitHub support website development and project file management.

8.2 Disadvantages

1. **Dependence on Data Quality:** Inaccurate, incomplete, or inconsistent input data may affect the reliability of the visualisations.
2. **Limited Real-World Applicability:** The project uses synthetic healthcare data, so its findings may not represent actual patient populations.
3. **No Clinical Diagnosis:** The dashboard is intended for data exploration and cannot diagnose diseases or replace professional medical advice.
4. **Internet Requirement:** Access to the published Tableau Public dashboard and online project resources requires an internet connection.

5. **Limited Predictive Capabilities:** The current project primarily focuses on descriptive analytics and visualisation rather than validated disease prediction.
6. **Learning Requirements:** Users may need basic knowledge of Tableau and data interpretation to understand some visualisations.
7. **Manual Data Updates:** Updated information may require changes to the dataset and a refresh of the relevant Tableau workbook, depending on the configuration.
8. **Limited Integration:** The current implementation does not necessarily provide direct integration with hospital databases or electronic health record systems.

9. CONCLUSION

The Health360 – Smart Healthcare Analytics: Patient Risk & Lifestyle Insights project demonstrates how data analytics and visualisation tools can be used to organise and explore healthcare information. The project uses a synthetic healthcare dataset and Tableau Desktop to create interactive dashboards containing KPI cards, bar charts, pie charts, heatmaps, and filters.

The dashboard presents an overview of patient demographics, exercise levels, average blood pressure, BMI categories, and recorded indicators for diabetes, heart disease, and stroke. The interactive filters enable users to examine different categories of the dataset, while the Tableau Story provides a structured way to present the visualisations and their observations.

The project also incorporates a Flask-based website developed using Python, HTML, and CSS, with GitHub used to manage project files and documentation. This combination demonstrates the practical application of data preparation, business intelligence, data visualisation, and web development.

Overall, Health360 achieves its educational objective of transforming raw healthcare data into a more accessible and understandable format. The project provides a foundation for further development in healthcare analytics. However, because it uses synthetic data and descriptive visualisations, its findings should not be considered clinical evidence or used for medical decision-making.

10. FUTURE SCOPE

The Health360 project can be enhanced in the following ways:

1. **Integration with Real Healthcare Data:** With appropriate authorisation, privacy protection, and data security, the project could be adapted to analyse anonymised real-world healthcare datasets.
2. **Predictive Analytics:** Machine learning models could be explored to estimate health risks, provided they are trained, validated, and evaluated using suitable data.
3. **Advanced Visualisations:** Additional charts could be developed to examine relationships between demographic characteristics, lifestyle factors, and health indicators.
4. **Real-Time Data Updates:** Automated data refresh mechanisms could be introduced to keep dashboards updated when new data becomes available.
5. **Improved Website Features:** The Flask website could be enhanced with improved navigation, a project overview, documentation, and easier access to dashboard resources.
6. **Enhanced User Experience:** Dashboard layouts, chart labels, colours, and filter placement could be improved to make the application more accessible and user-friendly.
7. **Expanded Reporting:** Downloadable reports and summaries could be added to support project presentations and analytical reviews.
8. **Security and Privacy Improvements:** If the system is adapted for real patient data, appropriate access controls, encryption, privacy safeguards, and compliance with applicable regulations would be necessary.

9. **Model Evaluation:** Any future predictive features could be tested for accuracy, reliability, fairness, and limitations before being considered for practical use.

These improvements could extend Health360 from an educational analytics demonstration into a more comprehensive healthcare data analysis platform.

11. APPENDIX

The appendix contains supporting information, screenshots, and technical resources related to the development and implementation of the Health360 project.

11.1 Dataset

- Dataset name: smart_healthcare_dataset.csv
- Purpose: To provide patient information for healthcare analytics and visualisation.
- Important fields: Age, Age Group, Gender, BMI, BMI Category, Blood Pressure, Exercise Level, Alcohol, Smoking, Diabetes, Heart Disease, and Stroke.

11.2 Software and Technologies Used

Technology	Purpose
Python	Backend programming and website development
Flask	Web application framework
HTML	Website structure

CSS	Website styling
Tableau Desktop	Data analysis and dashboard development
Tableau Public	Publishing and sharing visualisations
GitHub	Source code and project documentation management
CSV	Storage of the healthcare dataset

11.3 Project Resources

- GitHub repository: Health360 – Healthcare Analytics
- Dashboard: https://public.tableau.com/views/Health360Project-FINAL/Dashboard1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- Story : https://public.tableau.com/views/HealthcarePerformanceStory/Story1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- Dataset: https://drive.google.com/file/d/1g5N6hg1g5fw4BPdNEmw6a0QLjmui0Q7F/view?usp=drive_link
- Project Demo Link: https://drive.google.com/file/d/1eQsDEK5MhkDhScfS7tEfATmZtizWClrQ/view?usp=sharing